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PALEOMAGNETISM OF THE LOWER JURASSIC COPPER MOUNTAIN INTRUSIONS AND THE GEOTECTONICS OF TERRANE I, BRITISH COLUMBIA

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Abstract. The 198 Ma Lower Jurassic Copper Mountain intrusions outcrop in the Quesnellia Subterrane of Terrane I in the Canadian Cordillera. Thermal demagnetization at 550 ± 10°C defines a single stable remanence component of 25.9°, 41.2° (α_{95}=3.6°) in 11 reliable sites. The resulting pole position of 11.8°E, 57.3°N (δ_p = 2.7°, δ_m = 4.4°) is discordant for the North American craton and reflects 39 ± 6° of clockwise rotation and 9 ± 5° of northward translation. Combined with previous paleomagnetic and geologic data, a consistent model for the geotectonic evolution of Quesnellia and Terrane I emerges.

Introduction

Autochthonous composite Terrane I, essentially the Intermontane Belt of the Canadian Cordillera, has been subdivided into four subterranes [Monger et al., 1982]. Three paleomagnetic results are available for the southernmost Quesnellia Subterrane (Figure 1). The oldest rocks are the ~218 Ma mid-Triassic Nicola volcanics which retain a stable high temperature remanence component that suggests ~45° of clockwise rotation and ~14° of northward translation [Symons, 1984]. They are intruded by the ~202 Ma Lower Jurassic Guichon batholith that indicates ~43° of clockwise rotation and ~13° of northward translation [Symons, 1983a], and by the ~198 Ma Copper Mountain intrusions that indicate ~45° of clockwise rotation also but ~6° of southward translation [Symons, 1973]. The indicated southward translation conflicts with the paleomagnetic data for the Stikinia Subterrane which call for ~14° of northward translation assuming the two subterranes moved together [Monger and Irving, 1980; Symons, 1983b]. It has also created a problem for recent geotectonic models of Cordilleran evolution [e.g., Beck, 1980; Beck et al., 1981]. Presuming that the specimens used in the original study were insufficiently demagnetized by the alternating field (AF) technique employed, the collection was recored and subjected to further thermal demagnetization.

Geology

The Copper Mountain intrusions were emplaced into the Wolf Creek Formation of the Nicola Group (Figure 2). This group is comprised of subaerial and submarine alkalic to calc-alkalic andesites and basalts with minor volcanoclastic tuffs, clastic sediments and reef limestones. They are massive unfoliated rocks of the prehnite-pumpellyite metamorphic facies except near later intrusions, and they have low to moderate dips except near major later faults [Preto, 1972, 1979]. Late Karnian to mid Norian fossils indicate an age of 218 ± 4 Ma [Odin, 1982] for primary deposition which agrees with minimum K-Ar mineral ages of 207 ± 5 and 204 ± 5 Ma for the crosscutting Allison Lake Pluton using the Steiger and Jager [1977] decay constant.

The Copper Mountain intrusions include four bodies: i) the Copper Mountain stock (~12 km^2) has an outer diorite phase with gabbroic zones that grades inwards through monzonite to a perthositic pegmatite core; ii) the Voight stock (~5 km^2) is composed of a similar diorite; iii) the Lost Horse intrusion (~4 km^2) contains albitized and sausseritized diorite, monzonite and syenite cut by latite to trachyte dikes; and, iv) the Smelter Lake stock comprised of diorite only (Montgomery, 1967). The four bodies give Lower Jurassic minimum K-Ar mineral ages between 191 ± 7 and 203 ± 9 Ma [Sinclair and White, 1968; Preto et al., 1971; Preto, 1972]. They contain massive nonfoliated rocks except on the west side of the Lost Horse intrusion where extensive shearing and alteration are found near the Boundary Fault.

The Copper Mountain intrusions are cut by the 102 ± 4 Ma Verde Creek quartz monzonite and by the northerly-trending Mine Dykes of felsite and quartz/feldspar porphyry. The latter are believed to be related to the unconformably overlying Eocene Princeton Group of basic volcanics and clastic sediments [Preto, 1972]. Deformation consists of broad N-NW trending folds and extensive faulting especially of the Nicola Group. The entire zone is thought to be part of an ancient rift system that has been active since early Triassic time [Preto, 1979].

Methods

In the original study 212 specimens were collected from 21 sites (Figure 2). After excluding specimens that had been AF step demagnetized or glued together, 155 remained for thermal study. These were analyzed inside a shielded room using a Schonstedt SSM-1A spinner magnetometer and TSD thermal demagnetizer. After measuring their natural remanent magnetization, 21 specimens or one per site were demagnetized in 13 steps to 680°C. Their "knee-shaped" intensity decay plots (Figure 3a) indicate that magnetite with Curie temperatures close to 580°C is the main remanence carrier. Their Zijderveld plots (Figure 3b) indicate that only a single stable component is present above 520°C. Therefore, the remaining specimens were demagnetized at either 540°C (sites 3-6, 8, 9, 11, 16, 19, and 20) or 560°C (remaining sites). Only magnetically homogeneous specimens with a within-specimen angular standard deviation (ASD) of <25° were used for statistical analysis [Stupavsky and Symons, 1982].

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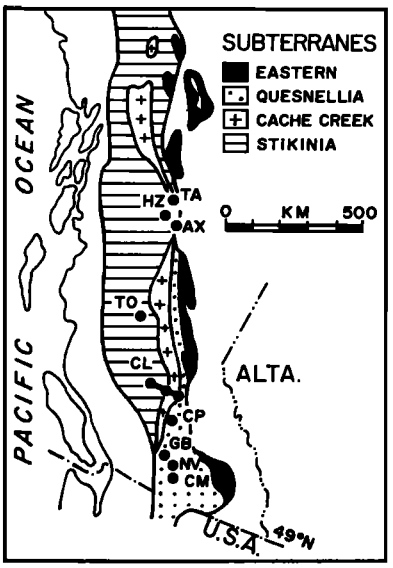


Fig. 1. Subterranes of Terrane I, British Columbia, after Monger et al. [1982] with the locations of paleomagnetic studies of the Copper Mountain intrusions (CM), Nicola volcanics (NV), Guichon batholith (GB), Cariboo lavas (CL) and plugs (CP), Topley intrusions (TO), Axelgold intrusion (AX), and Hazelton (HZ) and Takla (TA) volcanics.

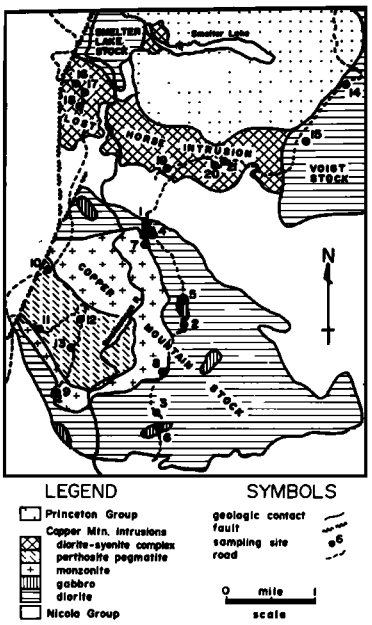


Fig. 2. Geology of the Copper Mountain intrusions after Montgomery [1967] with sampling site locations.

The NRM directions were plotted on an equal-area stereonet and contoured using a 1% smoothing area (Figure 4a). They yielded an elongate anomaly with a peak density of >19% (>>99.9% confidence) that includes the present Earth's magnetic field (EMF) direction, thereby indicating that viscous components are probably present.

After 550 ± 10°C thermal cleaning, the remanence directions from sites 16, 17 and 18 in the Lost Horse intrusion were divergent, probably because of rotations associated with the adjacent Boundary Fault (Table 1, Group III). A further 27 specimens did not meet the ASD criterium, leaving a population of 103 specimen directions from the 18 sites. These defined an "ideal" circular anomaly (Figure 4b) with a peak density of >25% that is indicative of the isolation of a single stable component of high statistical significance. Its mean direction was obtained by first averaging the specimen directions within the anomaly to obtain the site mean directions (Fisher, 1953). Four sites (3, 4, 5 and 21) with very weak remanence intensity and poor stability do not contribute to the anomaly. A further four sites (Table 1, Group II) are represented by only one or two specimens, and hence deemed unreliable even though their inclusion does not significantly change the mean of 25.9°, 41.2° (3.6°) (declination, inclination (radius of 95% confidence)) computed from the 11 reliable site means (Group I).

Discussion

The pole position for the Copper Mountain intrusions is located at 11.8°E, 57.3°N (δ_p = 2.7°, δ_m = 4.4°) using a mean site location of 239.5°E, 49.3°N. The 198 Ma reference pole position for cratonic North America is 94°E, 70°N (6°) [Irving, 1979]. Comparison of the two poles indicates that the Copper Mountain intrusions have undergone 39 ± 6° of clockwise rotation and 9 ± 5° of northward paleolatitudinal displacement since emplacement. The former is the same but more precise than the original estimate of 40 ± 10° [Symons, 1973], but the latter is significantly different from the original value of 6 ± 7° of southward translation. The Lower Jurassic Copper Mountain intrusions and Guichon batholith and the mid-Triassic Nicola volcanics (A component) represent an exposure length of 200 km in Quesnellia. All three define clockwise rotations of 39 ± 6°, 43 ± 7° and ~45°

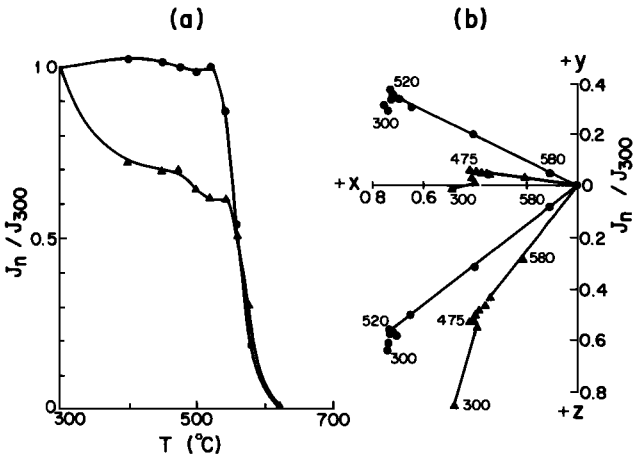


Fig. 3. Thermal step demagnetization data for specimens from sites 13 (circles) and 15 (triangles) showing: a) the normalized intensity decay curves; and b) the Zijderveld plots.

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respectively [Symons, 1983a and 1984] giving an average of ~42 ± 6°, and northward translations of 9 ± 5°, 13 ± 6° and 5° respectively giving an average of ~11 ± 5°. Combined with previous paleomagnetic data for Terrane I, a relatively clear geotectonic evolution emerges.

By Lower Jurassic time Terrane I had been assembled because the Quesnellia units and the Upper Triassic Takla volcanics in Stikinia [Monger and Irving, 1980] all have paleolatitudes that ~13° to the south of their present positions relative to the craton. Noting that mid-Jurassic to mid-Cretaceous units in Stikinia, i.e. the Topley intrusions [Symons, 1983b], Hazelton volcanics and Axelgold intrusion [Monger and Irving, 1980], have paleolatitudes indicative of ~14 ± 4° of northward motion, and that North America was moving westward from the mid-Atlantic Ridge, it appears that Terrane I was accreted to and moving with North America during Lower Jurassic to mid-Cretaceous time. Geological evidence suggests that accretion occurred by mid-Jurassic time [Monger et al., 1982], possibly associated with subduction of crust originating in the ancestral East Pacific Rise prior to its being overrun by North America.

After mid-Cretaceous time, Terrane I began sliding northward along the west margin of the North American craton. Monger et al. [1982] document the geologic evidence for major right-lateral strike-slip faults that could accommodate up to ~1400 km of displacement. The cause of this motion is uncertain. Possibly North America overrode a plate boundary so that northward motion of the Kula plate by rifting or of the Pacific plate by transform faulting was involved [e.g., Atwater, 1970; Stone et al., 1982]. It could also be associated with oblique subduction which led to the formation of the Coast Plutonic Complex on the west side of Terrane I. The complex contains plutons with K-Ar dates ranging from ~135 to ~45 Ma [Hutchison, 1970] as well as paleomagnetic data [e.g., Symons, 1977] indicative of northward motion. Further, the most likely age for the B remanence component in the Nicola volcanics of Quesnellia is mid-Eocene, and it suggests that ~6° of the northward motion predates and ~7° postdates this time [Symons, 1984].

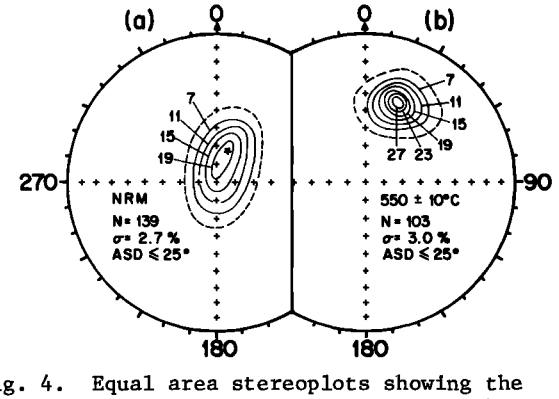


Fig. 4. Equal area stereoplots showing the population density contours for the: a) NRM, and b) 550 ± 10°C thermal cleaning. N is the number of population vectors and σ is the 95% confidence limit.

Table 1. Site Mean Remanence Data after 550 ± 10°C Thermal Cleaning

	Number				Mean Remanence Direction					
	a	b	c	d	R	D°	I°	k	α_{95}°	G
1*	11	1	3	7	6.8587	28.7	40.2	43	9.4	I
2*	5	1	2	2	1.9716	25.9	37.4	35	43.5	II
3	4	2	2	0						
4	8	5	3	0						
5	5	4	1	0						
6	7	5	1	1		45.6	57.5			II
7*	7	0	2	5	4.9404	27.6	41.0	67	9.4	I
8	5	2	2	1		23.8	44.9			II
9	7	0	3	4	3.8408	32.5	44.9	19	21.7	I
10	6	0	0	6	5.8753	22.0	40.4	40	10.7	I
11	8	2	2	4	3.8964	30.6	43.2	29	17.3	I
12	6	0	0	6	5.9444	21.9	40.9	90	7.1	I
13	13	0	0	13	12.9413	22.0	42.3	205	2.9	I
14	10	0	3	7	6.7021	31.6	40.5	20	13.8	I
15	6	0	1	5	4.9601	13.0	46.7	100	7.7	I
16	5	0	0	5	4.9593	35.9	64.4	98	7.8	III
17	10	0	2	8	7.9216	8.4	56.5	89	5.9	III
18	10	5	1	4	3.7848	12.7	50.9	14	25.5	III
19	11	0	0	11	10.8537	27.2	43.4	68	5.6	I
20	5	1	0	4	3.8439	26.7	27.8	19	21.5	I
21	6	4	2	0						
Group Averaged										
I				11	10.9376	25.9	41.2	160	3.6	
I + II				14	13.8731	26.7	42.4	102	3.9	
III				3	2.9692	17.3	57.8	65	15.4	

Notes: The * denotes reversed sites. Number of specimens: a) tested; b) having ASD 25°; c) outside 95% confidence anomaly; and, d) incorporated in mean. R is the resultant vector, D and I are the declination and inclination, k is the precision parameter, and α_{95} is the radius of 95% confidence [Fisher, 1953]. G is the group with: I) reliable, II) low precision, and III) Lost Horse Intrusion rotated.

During the northward motion, portions of Terrane I were rotated about vertical axes. The Topley intrusions in the interior of Stikinia are not rotated, supporting the notion that Terrane I was adjacent to the craton since mid-Jurassic time. However, the eastern marginal blocks of Terrane I containing the Takla volcanics, Hazelton volcanics and Axelgold intrusions have substantial clockwise and counterclockwise rotations in excess of 100° in some cases. These are attributed to "ball-bearing" type rotations [Beck, 1976] as Terrane I moved northward against the Omineca Complex to the east [Symons, 1984]. Quesnellia as a whole was also rotated by ~43° clockwise during the translation, with the Nicola B component implying that ~28° of the rotation predates and ~15° postdates mid-Eocene time. Directional changes in the sedimentary and structural trends between Quesnellia and Stikinia [Symons, 1971] and paleomagnetically determined rotations observed in the Cenozoic rocks of Washington and Oregon to the immediate southwest [e.g., Simpson and Cox, 1977] are consistent with

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a ~43° clockwise rotation for Quesnellia.

Paleomagnetic poles from the Cariboo lavas [Symons, 1969a] and plugs [Symons, 1969b] which overlie and intrude both Stikinia and Quesnellia are concordant and show that translations and rotations in Terrane I terminated by mid-Miocene time.

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